PRIYA AGARWAL

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Results oriented Data Scientist with 14+ years' proficiency in data synthesis, advanced predictive statistical analysis, automation and translating complex analyses into impactful business decisions.

Proficient in coding language (R), statistical regression models (Logistic, Cox), data query language (R-SQL), leveraging data visualization tools (R-ggplot2), extracting insights and creating dynamic reports that combine narrative text with embedded R code (R-Markdown) across multiple projects. Proven cross function collaborator to drive data-informed business outcomes.

SOFTWARE PROFICENCY

- **Software's:** SAS, R (R-Studio, R-Markdown), Phoenix WinNonlin
- **R-Packages:** ggplot2, haven, sas7bdat, gridExtra, lubridate, scales, flextable, tidyverse, readxl, rvest, survival, coxme
- **Computer Proficiency:** SQL, Google documents, MS Office, MS Access, Visual Basic

PROFESSIONAL EXPERIENCE

Sr. Principal Scientific Researcher/Scientist 4; Genentech | December 2010 – current | South San Francisco

Statistical data analysis:

- Informed strategic product improvement decisions **uncovering critical insights** supporting product improvement (drug development, dosing strategies and drug label) by leveraging R programming skills to format, re-structure and **analyze complex datasets** from multiple different sources (SAS, csv, excel).
- **Reduced analysis time by 60%** and accelerated high-stakes project timelines by implementing efficient coding strategies, including custom R functions to **automate** data visualization and statistical analyses across multiple projects.
- Achieved significant cost savings in future clinical trials study design by applying **Cox regression models (coxme)** on pooled data (~12k patient records, across 11 different studies), generating **forest plots**, estimating clinical event probabilities and assessing the association of a categorical variable with efficacy within various subgroups.
- Utilized **Logistic regression models** on pooled data to assess the relationship between two variables across various clinical trials to understand the ethnic differences in the population data crucial for personalized medicine approaches.
- Conducted **statistical analyses (t-test)** and developed diverse data visualizations with **ggplot2** in R to quantify and present differences in drug exposure between study arms, informing critical decisions for clinical trials.

Data processing and validation:

- Ensured the accuracy of all analytical outputs through severe **data quality validation and reconciliation** processes.
- Managed **multiple studies analysis by prioritizing** them and working independently in a time sensitive environment.
- Experience in testing internal AI/ML tool for data visualization, coding and analysis.
- Solved clinical pharmacology problems by integrating essential elements, including the end-to-end process of streamlining processes, and ensuring robust data tracking and analysis dataset delivery.

Cross-functional collaboration:

- Partnered with stakeholders to articulate data-driven stories via **R-markdown reports**, fostering insights and discussions.
- Collaborated with Roche's Basel team during a short rotation project and provided critical study insights on pooled data.
- **Cross-functionally collaborated** the writing of important health authority documents critical for drug approval processes.
- **Expedited data analysis** and improved team efficiency through early data access and performing the front-loading steps.

Programmer Analyst
Axistat | April 2010 – December 2010 | San Francisco

- Developed efficient programming, tested and validated SAS code for creating tables, listings and figures for the reports.
- Assisted Statisticians with maintaining analysis database specifications for ADAM datasets.

Service Associate, Financial Statistics and Information Department
American Red Cross | April 2007 – August 2008 | San Francisco

- Analyzed, reported and audited the use of Client Assistance Cards (CAC) assigned to clients during a Disaster Relief operation using MS Excel and the in-house CAC system.
- Trained Bay Area residents on earthquake preparedness and safety best practices.

Consultant, Marketing Team
Genpact (formerly GE Capital International Services) | July 2003 – December 2006 | India

- Extracted, automated and analyzed financial reports using SAS macros and SQL for Genworth Insurance's U.S. business.
- Analyzed year-wise distribution of mailers and responders across the country using MS Access.
- Developed arrays in SAS and worked in cross-functional team to test and analyzed files from different sources.

EDUCATION AND CERTIFCATION

- Master in Science, Biostatistics, California State University, Hayward, USA (2008-2010)
- Master in Science, Applied Statistics and Informatics, Indian Institute of Technology, Mumbai, India (2001-2003)
- Bachelor of Adminitsration Mathematics, Lady Shri Ram College, Delhi University, New Delhi, India (1998-2001)
- Certifications: Advanced SAS, Base SAS certification

POSTERS AND PUBLICATIONS

- Co-authored 21 posters/abstracts (**4 as a first author**) at various conferences; ASCPT, ACCP, PAGE etc.
- Co-authored 17 manuscripts/papers (**4 as a second author**) for various journals.

List of first author posters and second author manuscripts:

1. Pharmacokinetics of Venetoclax in Combination with Rituximab in Patients with Relapsed/Refractory Chronic Lymphocytic Leukemia: Phase 3 MURANO Study. [**Agarwal, P.**, et al. (2018); ACCP]

2. Clinical Pharmacokinetics (PK) of polatuzumab vedotin (Pola) in combination with rituximab (R), obinutuzumab (G), cyclophosphamide (C), doxorubicin (H) and bendamustine (B) in 1L or Relapsed/Refractory (R/R) Non-Hodgkin's Lymphoma (NHL) patients. [**Agarwal, P.**, et al. (2017), ACCP]

3. Phase I study of trastuzumab emtansine in HER2-positive metastatic breast cancer patients with normal or reduced hepatic function [**Agarwal, P.**, et al. (2016); ASCPT]

4. Assessment of the potential for Drug-Drug interactions between Trastuzumab Emtansine (T-DM1) and CYP3A inhibitors or inducers and the impact on its PK and safety. [**Agarwal, P.**, et al. (2015); ASCPT]

5. Optimizing Early-stage Clinical Pharmacology Evaluation to Accelerate Clinical Development of Giredestrant in Advanced Breast Cancer. [Malhi, V., **Agarwal, P.**, et al. (2023); PMID: 38019116]

6. Relative Bioavailability and Food Effect Testing of a Pediatric-Friendly Oral Suspension of Alectinib in Healthy Volunteers Using Venipuncture and Capillary Microsampling. [Liu, S., **Agarwal, P.**, et al. (2023); PMID: 36978270]

7. Mitigating ipatasertib-induced glucose increase through dose and meal timing modifications. [Liu, S., **Agarwal, P.**, et al. (2023); PMID: 36197694]
8. A Phase I Pharmacokinetic Study of Trastuzumab Emtansine (T-DM1) in Patients with Human Epidermal Growth Factor Receptor 2-Positive Metastatic Breast Cancer and Normal or Reduced Hepatic Function. [Li Chunze., **Agarwal, P.**, et al. (2017); PMID 27995530]